CURRICULUM VITAE

Name: Dr. Amit K Parikh

Address: **Residence:**

27, Safal Vivaan, Phase 2,

Nr.Gota Overbridge, S-G Highway,

Ognaj, Ahmedabad-382 481

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Mehsana Urban Institute of Sciences

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Email: principal.mubiob@ganpatuniversity.ac.in, amit.parikh.maths@gmail.com

Designation: Professor & Principal at Mehsana Urban Institute of Sciences, Ganpat University

Dean, Faculty of Science, Ganpat University

Date of Birth: 22/08/1972

Language Skills: Ability to communicate in English, Hindi and Gujarati

Educational Qualification:

Degree	University	Month &Year	Percentage
B.Sc. (Mathematics)	GujaratUniversity	May, 1992	78
M.Sc. (Mathematics)	GujaratUniversity	May, 1994	68
M.Phil. (Mathematics)	GujaratUniversity	July, 1998	77
Ph.D.	SVNIT, Surat	April, 2014	

M.Phil. Dissertation:

- **Title:** Study of Continuous Functions
- **Guide:** Prof. I. H. Sheth, Department of Mathematics, Gujarat University, Ahmedabad.

Ph.D. Thesis:

- **Title:** Analytical solution of nonlinear partial differential equations of some phenomena arising in fluid flow through porous media
- **Guide:** Prof. M.N.Mehta & Dr.V.H.Pradhan, SVNIT, Surat.

Membership:

1. Executive Committee member of Gujarat Ganit Mandal
2. Academic Council member of Ganpat University
3. Member of Board of Governors of Ganpat University
4. Life Member of Indian Society for Technical Education (ISTE), New Delhi. [LM 33017]
5. Life Member of Indian Society of Theoretical and Applied Mechanics (ISTAM) [L 676]
6. Life Member of Gujarat Ganit Mandal, Ahmedabad.
7. Life member of GAMS (Gwalior Academy of Mathematical Sciences)
8. Life Member of FATER (Forum for advanced training education and research academy of India)
9. Editorial Board member of International Journals IJIRMF & IJRCS run by "RESEARCH CULTURE SOCIETY".
10. Associate fellow member and Executive Council Member of "Gujarat Science Academy"
11. General Associate Member of The Mathematics Consortium

Experience: 25 years (Teaching, Research & Administration)

Sr.No.	Organization	Designation	Duration
1	C.U. Shah College of Engineering & Technology, Wadhwan city.	Lecturer	16/09/1999 to 19/09/2001 (i.e. 2 Years)
2	K.B. Shah Science College, Wadhwan.	Visiting Lecturer	16/09/1999 to 15/09/2001 (i.e. 2 Years)
3	Nirma Institute of Diploma Studies, Ahmedabad.	Lecturer (Sr. Scale)	20/09/2001 to 20/06/2007 (i.e 5 years & 9 months)
4	Nirma University, Institute of Technology, , Ahmedabad.	Assistant Professor	21/06/2007 to 21/05/2014 (i.e. 6 years &11 months)
5	C U Shah University, Faculty of Engg. & Tech., Wadhwan city	Professor	22/05/2014 to 21/12/2015 (i.e 1 year & 7 months)
6	Mehsana Urban Institute of Sciences, Ganpat University, Kherva	Professor & Principal	22/12/2015 onwards

As a Ph.D Supervisor

1. Ms.Sweta H. Shah (14SC702003) completed her Ph.D thesis titled "Structured Group of Matrices of different classes and Cocentroidal Matrices in JS Metric Space" from C.U.Shah University, Wadhwan in December 2017.
2. Mr.Himesh Arvindbhai Patel (PHD1308001) completed his Ph.D thesis titled "Performance Analysis of Rough bearing system under Newtonian fluid / Non Newtonian fluid" from C.U.Shah University, Wadhwan in October 2018.
3. Ms.Vaishali Achesariya (14SC702001) completed her Ph.D thesis titled "Study on some classes of square matrices along with operator matrices and their properties" from C.U.Shah University, Wadhwan in May 2019.
4. Mr.Pragnesh Makwana (15SC702003) completed his Ph.D thesis titled "**Mathematical Modeling of some phenomena arising in Fluid Flow through porous media** " from C.U.Shah University, Wadhwan in October 2021.
5. Mr.Jishan Shaikh (**16146061002)** completed his Ph.D thesis titled "**Numerical Solution of Non-Linear Partial Differential Equation Arising In Fluid Flow through Porous Medium**" from Ganpat University, Kherva in December 2021.
6. Ms.Disha Shah (17146061002) completed her Ph.D thesis titled **"Classical Solution of some non-linear partial differential equations"** from Ganpat University, Kherva in December 2021.
7. Ms.Dhara Patel (17146061001) completed her Ph.D thesis titled **"Asymptotic Behaviour of Nonlinear Fluid Flow Problems in Porous Media"** from Ganpat University, Kherva in March 2023.
8. Currently five students are pursuing Ph.D at Ganpat University, out of them one student have completed Pre-Synopsis.

LIST OF NATIONAL/ INTERNATIONAL PUBLICATIONS

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1. Parikh A.K., Mehta M.N., Pradhan V.H., *The atmospheric pressure in dry region and velocity of infiltered water in ground water infiltration phenomenon.*, **Int. J. of. Appl. Math and Mech,** 7(13): 77 – 90, (2011). ISSN 0973-0184.
2. Parikh A.K., Mehta M.N., Pradhan V.H., *Transcendental solution of Fokker-Planck equation of vertical ground water recharge in unsaturated homogeneous porous media,* Journal of **IJERA,** 1(4):1904-1911, (2011). ISSN 2248-9622 (Online)
3. Parikh A.K., Mehta M.N., Pradhan V.H., *A Numerical Solution of the flow of two immiscible fluids through porous media with mean pressure*, **Bio-IT(IJACM)**, 3(2):237-244, (2012).[IFV: 2.95], ISSN 2230–9624
4. Parikh A.K., Mehta M.N., Pradhan V.H., *Generalised separable solution of double phase flow through homogeneous porous medium in vertical downward direction due to difference in viscosity*, **Applications and ApplEVIied Mathematics: An International Journal**,8(1): 305- 317, (2013). ISSN 1932-9466
5. Parikh A.K., Mehta M.N., Pradhan V.H., *Mathematical model and analysis of counter-current imbibition in vertical downward homogeneous porous media*, **British Journal of Mathematics & Computer Science**,3(4): 478-489, (2013), ISSN **2231-0851**
6. Parikh A.K., Mehta M.N., Pradhan V.H., *Generalised separable solution of counter-current imbibition phenomenon in homogeneous porous medium in horizontal direction*, **The International Journal of Engineering and Science**, 2(1):220-226, (2013), ISSN 2319–1813
7. Parikh A.K., Mehta M.N., Pradhan V.H., *Generalized separable solution of double phase flow through homogeneous porous medium in horizontal direction due to difference in viscosity*, **J. Environ. Res. Develop**,8(1): 109-114, (2013) [IF 1.613], ISSN 0973-6921

8. Parikh A.K., Mehta M.N., Pradhan V.H., *Transcendental Solution of Infiltration Phenomenon with External Effect, Int. J. of. Appl. Math and Mech., 9 (19): 60-70, (2013). ISSN 0973-0184*

9. Parikh A.K., Mehta M.N., Pradhan V.H., *A Mathematical Modeling and Analysis of Fingero-Imbibition phenomenon in vertical downward cylindrical homogeneous porous matrix, Proceedings of NUiCONE in IEEE journal., ISSN 978-1-4799-0726-7*

10. Parikh A.K., Mehta M.N., Pradhan V.H., *Mathematical Modeling and Analysis of Fingero-Imbibition phenomenon in homogeneous porous medium with magnetic field effect in vertical downward direction, IJLTEMAS, III (X): 17-23, (2014). ISSN 2278- 2540*

11. D. A. Shah, A.K.Parikh, *Solution of One Dimensional Wave Equation using Laplace Transform, IJSR, 4(2): 2264-2266, (2015). [ISSN (online) 2319-7064, IF:4.438]*

12. A.K.Parikh, *Generalized Separable Method to solve Second order non linear Partial Differential Equations,* **IJAR**, 5(3): 263 – 269, (March 2015), [ISSN: 2249-555X IF: 2.1652].

13. A.K.Parikh, *Numerical Solution of Boussinesq's Equation in Ground Water Infilteration Phenomenon by Differential Quadrature Method,* **PIJR**, 4(6):165-167, (June 2015), [ISSN: 2250-1991] IF: 3.4163 **UGC Approved: 47432**

14. A.K.Parikh, *Mathematical model of Fingero-Imbibition phenomenon arises during secondary oil recovery process,* **Mathematics Today**, Vol – 30, 1-13 (May 2015), [ISSN: 0976-3228], **UGC Approved: 47789**

15. D. A. Shah, A.K.Parikh, *Functional Separable Solution of Nonstationary Heat Equation with a Nonlinear Source,* The Proceedings of International Conference on Emerging Trends in Scientific Research (ICETSR-2015), Page no: 196-199 (Dec 2015), [ISBN: 978-2-642-24819-9]

16. R.H.Daslania, A.K.Parikh, *Mathematical model of two dimensional wave equation in rectangular membrane*, The Proceedings of International Conference on Emerging Trends in

Scientific Research (ICETSR-2015), Page no: 219-222 (Dec 2015), [ISBN: 978-2-642-24819-9]

17. *Pragneshkumar Makwana, Amit K Parikh, Submultiplicativity and Different norms on* $M_{m,n}(C)$*,* The Proceedings of International Conference on Emerging Trends in Scientific Research (ICETSR-2015), Page no: 241-247 (Dec 2015), [ISBN: 978-2-642-24819-9]
18. Pradip J Jha, Amit K Parikh, Published an article titled "Exploring the Exciting Era" in Souvenir of 53rd Gujarat Ganit Mandal Adhivation organized by Mehsana Urban Institute of Sciences, Ganpat University during 23-25 December 2016, Page Nos. 76 – 78.
19. A.K.Desai, Amit K Parikh, Published an article titled "Use of *Mathematica* in handling Infinite series, Integrals and Products involving Pi (π) " in Souvenir of 53rd Gujarat Ganit Mandal Adhivation organized by Mehsana Urban Institute of Sciences, Ganpat University during 23-25 December 2016, Page Nos. 79 – 83.
20. A.K.Parikh*, Mathematical model of Instability phenomena in Homogeneous porous medium in vertical downward direction,* International Journal For Innovative Research In Multidisciplinary field (IJIRMF), No. 211 – 218, Vol - 3, Issue - 1, Jan- 2017, ISSN – 2455 – 0620. [IF: 6.7191 Index Copernicus] UGC - 47793
21. Sweta Shah, Pradeep Jha, Amit Parikh*, Classification of operator matrices and their application,* **International Journal of Advanced Science and Research***, 2(3):70-75 (July 2017),* ISSN: 2455-4227, Impact Factor: RJIF 5.12
22. Vaishali Achesariya, Pradeep Jha, Amit Parikh *Cocentroidal Matrices of Class 1 and Class 3,* **International Journal of Advanced Science and Research***, 2(3):66-69 (July 2017),* ISSN: 2455-4227, Impact Factor: RJIF 5.12
23. *Approximate Solution of Nonlinear Diffusion Equation Using Power Series Method (PSM),* International Journal of Current Advanced Research ISSN: O: 2319-6475, ISSN: P: 2319-6505, Impact Factor: SJIF: 5.995, Volume 6; Issue 10; October 2017; Page No. 6374-6377. UGC Approved: 43892

24. *Solution of one dimensional time dependent Reaction-Diffusion equation by Power Series Method (PSM),*Conference Proceedings of **3rd International Conference on Emerging Global Trends in Humanities, Science, Technology, Commerce & Management Education (ICEGTE 2018)** during 23-24 March 2018, organized by Chirist Institute of Management and Chirist College, Rajkot, Bharti Publication, ISBN: 978-93-86608-48-2, Page No. 183-186.

25. *Mathematical modeling and analysis of ground water infiltration in unsaturated porous media by using power series method (PSM),* International Journal of Recent Scientific Research, ISSN: 0976-3031, Impact Factor: 7.383, Vol. 9, Issue, 4(F), pp. 25918-25922, April, 2018. **UGC Approved: 46629**

26. *Solution of Nonlinear Partial Differential Equation Using Generalized Functional Separable Method*, Mathematics Today, Vol – 34 A, (April 2018), [ISSN: 0976-3228, e-ISSN: 2455 - 9601], Page no:32-39, **UGC Approved: 47789**

27. *Solution of One Dimensional Telephone Equation Using Method of Separation of Variables,* Mathematics Today, Vol – 34 A, (April 2018), [ISSN: 0976-3228, e-ISSN: 2455 - 9601], Page no:157 – 162, **UGC Approved: 47789**

28. *Approximate solution of Nonlinear Second Order Diffusion Equation arising in Ground Water Inltration Phenomenon using Power Series Method (PSM),* Mathematics Today, Vol – 34 A, (April 2018), [ISSN: 0976-3228, e-ISSN: 2455 - 9601], Page no:195 – 201. **UGC Approved:47789**

29. *Effect of Longitudinal Surface Roughness on the Performance of a Shliomis Model Based Ferrofluid Annular Squeeze Film Considering Couple stresses Effect,* Mathematics Today, *Vol* – 34 A, (April 2018), [ISSN: 0976-3228, e-ISSN: 2455 - 9601], Page no:228 – 239. **UGC Approved: 47789**

30. Pragnesh Makwana, Amit Parikh, *Power Series Method for Solving Kolmogorov-Petrovsky-Piskunov Equation,* JETIR International Journal (ISSN: 2349-5162), Vol – 5, Issue – 7, Page no. 277 – 278. [Impact factor 5.87, Indexed in Thomson Reuters], **UGC Approved: 63975**

31. Disha Shah, Amit Parikh, *Classical Solution of Non-stationary Heat Equation with Nonlinear Logarithmic Source Using Generalized Functional Separable Method (GFSM),* Journal of Computer and Mathematical Sciences, Vol.9(10),1301-1306 October 2018 [ISSN:0976-5727 (Print) ISSN:2319-8133(Online)], **U.G.C. Approved Journal No.: 44720**, Impact factor : 4.655

    [DOI: 10.29055/jcms/871]

32. Disha Shah, Amit Parikh. *Functional Separable Method (FSM) and its Applications to Solve Nonlinear Partial Differential Equation,* International Journal for Research in Engineering Application & Management (IJREAM) ISSN : 2454-9150, Vol-05, Issue-01, April 2019, Page no. 444 – 448, **UGC Sr.No. 64077.**

33. Jishan Shaikh, Amit Parikh, *Application of Polynomial based Differential Quadrature Method in double phase (Oil-Water) flow problem during secondary oil recovery process,* Indian Journal of Applied Research (IJAR), Volume-9, Issue-5, May-2019, PRINT ISSN No 2249 -555X, Page No. 19-24, Impact factor: 5.397, **UGC Sr. No. 49333.**

34. Dhara T Patel, Amit K Parikh, *Approximate analytical solution of counter-current imbibitions phenomenon using Homotopy Perturbation Method* , International Journal of Research and Analytical Reviews, Approved by **UGC Journal No-43602**, volume-6 Special issue conference 2019.

35. Krisha Shah, Amit K Parikh, *Solution of Groundwater Infiltration Phenomenon in horizontal directionBy DTM and RDTM Method,* International Journal of Research and Analytical Reviews, Approved by **UGC Journal No-43602**, volume-6 Special issue conference 2019.

36. *Numerical Solution of Imbibition Phenomenon Using PDQM in Vertical Downward Direction,* IJISET - International Journal of Innovative Science, Engineering & Technology,

Vol. 7 Issue 3, March 2020, **ISSN (Online) 2348 – 7968 | Impact Factor 9) – 6.248 [**UGC Approved SI. No. 4798 & 48287, Thomson Reuters Reseacher ID indexed Journal: Reseacher ID: O-3517-2016]

37. *Mathematical Model of Co-Current Imbibition in Homogeneous Porous medium in vertical downward direction*, Journal of Xidian University, ISSN No:1001-2400, VOLUME 14, ISSUE 11, 2020, Pg no. 546 – 553 [Impact Factor: 5.4, UGC Approved, SCOPUS Indexed], November 2020. [DOI: **10.37896/jxu14.11/051**]

38. *Mathematical Solution of Fingering Phenomenon In Vertical Downward Direction Through Heterogeneous Porous Medium,* Advances in Mathematics: Scientific Journal 10 (2021), no.1, 483–496 ISSN: 1857-8365 (printed); 1857-8438 (electronic), Spec. Issue on ICIRPS-2020 [SCOPUS Indexed] (DOI: 10.37418/amsj.10.1.48)

39. *Numerical solution of Countercurrent Imbibition in Inclined Homogeneous Porous Medium by using Polynomial based Differential Quadrature Method,* PalArch's Journal of Archeology of Egypt / Egyptology, 17(9). ISSN 1567-214x, pp 8045-8054 [SCOPUS Indexed], 2020

40. *Shaikh Jishan, Parikh A.K., Numerical Solution of Counter-Current* Imbibition *Phenomenon In Homogeneous Porous Media Using Polynomial Base Differential Quadrature Method With Chebyshev-Gauss-Lobatto Grid Points*, Mathematical Modeling, Computational Intelligence Techniques and Renewable Energy (MMCITRE2020), Advances in Intelligent Systems and Computing, AISC vol 1287, pp 195-206, Springer, Singapore. (https://doi.org/10 .1007/978-981-15-9953-8_17), Feb 2021, Print ISBN978-981-15-9952-1, Online ISBN978-981-15-9953-8

41. *Patel D. T., Parikh A.K., A Series solution of Nonlinear Equation for Imbibition Phenomenon in One-Dimensional Heterogeneous Porous Medium*, Mathematical Modeling, Computational Intelligence Techniques and Renewable Energy (MMCITRE2020), Advances in Intelligent Systems and Computing, AISC vol 1287, pp 29 - 40, Springer, Singapore.

(https://doi.org/10.1007/978-981-15-9953-8_4), Feb 2021, Print ISBN978-981-15-9952-1, Online ISBN978-981-15-9953-8

42. *Solution of Instability Phenomenon in Homogeneous Porous Media with and Without Inclination by Differential Transform Method,* International Journal for Research in Engineering Application & Management (IJREAM) ISSN : 2454-9150 Vol-06, Issue-12, MAR 2021 (UGC Approved Sr.No. 64077, Impact Factor : 6.466)
43. *Solution of Boussinesq's Equation for Infiltration Phenomenon in Horizontal Direction by Differential Transform Method,* IJISET - International Journal of Innovative Science, Engineering & Technology, Vol. 8 Issue 5, May 2021 ISSN (Online) 2348 – 7968 | Impact Factor (2020) – 6.72. (UGC Approved SI. No. 4798 & 48287, Thomson Reuters Reseacher ID indexed Journal: Reseacher ID: O-3517-2016)
44. Dhara T Patel, Amit K Parikh, *Approximate Solution of Non-Linear Equation for Imbibition Phenomena in One Dimensional Fractured Heterogeneous Porous Medium,* International Journal for Research in Engineering Application & Management (IJREAM) ISSN: 2454-9150 Vol-07, Issue-02, MAY 2021, pp-546 – 551. (UGC Approved Sr.No 64077)
45. Disha Shah, Amit Parikh, *Mathematical Solution of Imbibition Phenomenon in Vertical Heterogeneous Porous Medium,* Advances in Mathematics: Scientific Journal 10 (2021), no.10, 3349–3362 ISSN: 1857-8365 (printed); 1857-8438 (electronic) https://doi.org/10.37418/amsj.10.10.6 (SCOPUS), Oct 2021
46. Shah Disha, Amit Parikh, *Mathematical Modelling of Moisture Content In Unsaturated Heterogeneous Soil and Its Solution by Using Method of Functional Separable,* Advances and Applications in Mathematical Sciences Volume 21, Issue 6, April 2022, Pages 2935-2947 © 2022 Mili Publications, India, ISSN - 0974-6803 (UGC care listed, Web of Science).
47. Aruna Sharma, Amit Parikh, *Solution of Burger's Equation arising in Longitudinal Dispersion Phenomenon using Two-Dimensional Differential Transform Method,* Advances and

Applications in Mathematical Sciences Volume 21, Issue 6, April 2022, Pages 3161-3173 © 2022 Mili Publications, India, ISSN - 0974-6803 (UGC care listed, Web of Science).

48. Dhara T Patel, Amit K Parikh, *Homotopy Analysis Laplace Transform Method for Analytical Solution of Porous Medium Equation*, International Journal of Innovative Science and Research Technology (IJISRT), Volume-07, Issue-11, pp. 304-309, Nov2022, ISSN No:- 2456-2165

49. Shrikrishna Dasari, Amit Parikh, *Numerical Solution of 2D Inhomogeneous Helmholtz Equation using the Meshless Radial Basis Function Method* , IJISET - International Journal of Innovative Science, Engineering & Technology, Vol. 10 Issue 01, January 2023 ISSN (Online) 2348 – 7968 | Impact Factor – 6.72 (UGC care list SI No. 4798 & 48287)

50. Maitri R. Raval, Amit K. Parikh1, Bijal M. Yeolekar, *HIV-CM (cryptococcal meningitis) co-infection and control in immunocompromised individuals*, Eur. Chem. Bull. 2023, 12 (Special Issue7), 1463-1484, ISSN 2063-5346. (Scopus Indexed)

51. Rajeshri Prajapati , Amit Parikh , Pradeep Jha, *Special Graphs of Euler's Family* and Tracing Algorithm- (A New Approach),* The Ciência & Engenharia - Science & Engineering Journal ISSN: 0103-944X Volume 11 Issue 1, 2023 pp: 2243 – 2251 (Scopus Indexed)

52. Disha Shah, Amit Parikh, *Quantitative Analysis of Co-current Imbibition Phenomenon with Heterogeneity and Gravitational Effects*, Scope – Scopus and Cosmos Indexed Journal ISSN: 1177-5653(P), 1177-5661(O), Volume 13 (3) September 2023, pp 872 -882.

53. Shrikrishna Dasari, Amit Parikh, *Meshless Radial Basis Function Pseudo-Spectral Method for Solving Non-linear KdV Equation*, RGN Publications - Communications in Mathematics and Applications, Vol. 14, No. 3, pp. 1153–1160, 2023 ISSN 0975-8607 (online); 0976-5905 (print), Special Issue Recent Trends in Mathematics and Applications Proceedings of the International Conference of Gwalior Academy of Mathematical Sciences 2022, (DOI: 10.26713/cma.v14i3.2376)

54. Aruna Sharma, Amit Parikh, *Semi Analytic-Numerical Solution of Imbibition Phenomenon in Homogeneous Porous MediumUsing Hybrid Differential Transform Finite Difference Method,* RGN Publications - Communications in Mathematics and Applications, Vol. 14, No. 3, pp. 1199–1213, 2023 ISSN 0975-8607 (online); 0976-5905 (print), Special Issue Recent Trends in Mathematics and Applications Proceedings of the International Conference of Gwalior Academy of Mathematical Sciences 2022, (DOI: 10.26713/cma.v14i3.2394)
55. Pragnesh Makwana, Amit Parikh *Solution of nonlinear Burger's equation arising in longitudinal dispersion phenomena,* Results in Control and Optimization (2023), Elsevier https://doi.org/10.1016/j.rico.2023.100370
56. Krisha Shah, Amit Parikh, *Differential Transform Method applied on Groundwater Infiltration Phenomenon in horizontal direction with external effect (Input And Output),* REVISTA INVESTIGACION OPERACIONAL, Universidad de La Habana, VOL. 45, NO. 1, 25-32, 2024, ISSN: 0257-4306, E – ISSN: 2224-5405 (Scopus)
57. Pragnesh Makwana, Amit Parikh, *Application of Double Power Series Method for solving nonlinear Burger's equation arising in longitudinal dispersion phenomena,* REVISTA INVESTIGACION OPERACIONAL VOL. 45, NO. 1, 65-70, 2024, ISSN: 0257-4306, E – ISSN: 2224-5405 (Scopus)
58. Pragnesh Makwana, Amit Parikh, *Asymptotic solution of Fokker-Planck equation based on Darcy-Buckingham approach in unsaturated soil,* Journal of the Serbian Society for Computational Mechanics / Vol. 17 / No. 1, 2023 / pp 88-96 (WOS) (10.24874/jsscm.2023.17.01.06)

RESEARCH PAPER PRESENTATIONS IN CONFERENCE/SEMINAR

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1. *Solution of Boussinesq's equation in ground water infiltration phenomenon by perturbation technique*, **ISTAM: An International meet,** NIT Hamirpur, during 18/12/10 to 21/12/10.
2. *A Numerical Solution of the Flow of two immiscible fluids through Porous Media with mean pressure*, 16th Annual cum 2nd **International Conference of Gwalior Academy of Mathematical Sciences (GAMS),** during 22/09/11 to 24/09/11.
3. *Transcendental Solution of Vertical Ground water recharge in Unsaturated Porous Media*, **ISTAM: An International meet**, SVNIT, Surat during 19/12/11 to 21/12/11.
4. *Generalised separable solution of instability phenomenon in homogeneous porous medium*, **National Conference on Recent Trends in Mathematical Sciences and their Applications: NCRTMSA,** Mody Institute of Technology and Science (MITS), Lakshmangarh, Rajasthan, during 5/11/12 to 6/11/12.
5. *Generalised separable solution of double phase flow through homogeneous porous medium in horizontal direction due to difference in viscosity,* **5th International Congress of Environmental Research (ICER-12) at UMT, Terengganu (Malaysia)** during 22/11/12 to 24/11/12.
6. *Generalized separable solution of counter-current imbibition phenomenon in homogeneous porous medium in horizontal direction,* **International Conference in Mathematical Sciences (ICMS-2013)** at Government degree college, Haripur (Manali), during 8-9 March 2013.
7. *Mathematical Modeling and Analysis of Fingero-Imbibition phenomenon in vertical downward cylindrical homogeneous porous matrix,* **International Conference: NUiCONE** at Nirma University, Ahmedabad, during 28th - 30th November, 2013.
8. *Generalised separable method to solve nonlinear partial differential equation,* **TEQIP-II sponsored STTP** on 'Advanced Analytical & Numerical Techniques for Engineers and Scientists' during 3-7, March 2014 organized by SVNIT Surat.

9. *Generalised Separable Solution of Fingero-Imbibition phenomenon in vertical homogeneous porous matrix with magnetic effect,* 19th Annual cum 4th International Conference of Gwalior Academy of Mathematical Sciences(**GAMS**) on "**Advances in Mathematical Modeling to Real World Problems"** during 3-6, October 2014 organized by SVNIT Surat.

10. *Functional Separable Solution of Nonstationary Heat Equation with a Nonlinear Source,* International Conference on Emerging Trends in Scientific Research (ICETSR-2015) during 17-18 December 2015 organized by C.U.Shah College of Engineering and Technology, Wadhwan city.

11. *Approximate solution of Non-linear Second Order Diffusion Equation arising in Ground Water Infiltration Phenomenon using Power Series Method (PSM),* **22nd Annual cum 6th International Conference of the Gwalior Academy of Mathematical Sciences on 'Advances in Pure and Applied Mathematics' during 22-24 December 2017 organized by** Ganpat University, Kherva.

12. *Solution of one dimensional time dependent Reaction-Diffusion equation by Power Series Method (PSM),* **3rd International Conference on Emerging Global Trends in Humanities, Science, Technology, Commerce & Management Education (ICEGTE 2018)** during 23-24 March 2018, organized by Chirist Institute of Management and Chirist College, Rajkot.

13. *Mathematical modeling of physical phenomena arising during oil recovery process,* **68th International Conference on New Trends in Engineering, Science and Management** during 20-21 July 2018, organized by IOSRD at Visakhapatnam.

14. *Solution of nonlinear nonstationary wave equation using Clarkson-Kruskal direct method (CKDM),* **68th International Conference on New Trends in Engineering, Science and Management** during 20-21 July 2018, organized by IOSRD at Visakhapatnam.

15. *Analysis of One-Dimensional Groundwater Recharge by Spreading using Hybrid Differential Transform Finite Difference Method* , Eurasian Conference on 'Science, Engineering & Technological Innovations' during 20 -21 November 2021 Jointly organized by Kryvyi Rih

National University, Ukraine, Europe Automation, Computer Science and Technology Department Research Culture Society (online mode).

16. Presented a paper entitled "*Semi Analytic-Numerical Methods to solve OneDimensional Dispersion Phenomenon of miscible fluid flow through porous media*" in the 3rd International Conference on Mathematical Modeling, Computational Intelligence Techniques and Renewable Energy (MMCITRE - 2022) organized during March 04-06, 2022 conducted by University of Technology Sydney (UTS), Pandit Deendayal Energy University (PDEU), Universidad Autónoma de Baja California (UABC), Mexico, Universidad Católica de la Santísima Concepción (UCSC), Chile, Forum for Interdisciplinary Mathematics (FIM) and Victoria University, Melbourne, Australia (online mode).

17. Presented paper on "**Some Interesting Mathematical Expressions involving Pi(π) with Mathematica**" at **International Conference for Award Winners in Engineering, Science and Medicine (INSO - 2023)** held on **15-Apr-2023** in **Pondicherry**, India.

EXPERT TALK / INVITED TALK

1. Delivered an expert talk on 'Matrix' on 18th May 2010 at workshop on Advance Mathematics organized by Institute of Diploma Studies, Nirma University, Ahmedabad.
2. Chaired the session at International conference ICER-12 during 22-24 November 2012 organized at Universiti Malaysia Terengganu, Malaysia.
3. *Mathematical Model of Fingero-Imbibition phenomenon in vertical downward cylindrical homogeneous porous media with magnetic field effect,* as an expert talk at **International conference on Applications of Fluid Dynamics (ICAFD-2014)** during 21-23, July 2014 organized by S. V. University, Tirupati and University of Botswana, Gaborone, Botswana.
4. Delivered an expert talk on " Generalized separable method and its applications" at National conference on Recent trends in Mathematics and its applications (RTMA – 2014) on 19th Dec 2014 organized by Department of Mathematics, C.U.Shah University, Wadhwan city.

5. Chaired the session at International Conference on Emerging Trends in Scientific Research (ICETSR-2015) during 17-18 December 2015 organized by C.U.Shah College of Engineering and Technology, Wadhwan city.
6. Delivered an expert talk on "Application of Mathematics in Oil Recovery Process" at One day National Seminar on "Mathematics and its applications" on 19th November 2016 organized by Department of Mathematics & Computer Science, School of Technology, PDPU, Gandhinagar.
7. Delivered an expert talk on "Mathematical formulation of some physical phenomena occurring during secondary oil recovery process" at "53rd Annual Conference of Gujarat Ganit Mandal" during 23 – 25 December 2016 organized by Ganpat University, Mehsana Urban Institute of Sciences, Kherva.
8. Delivered an expert talk on "An extension of Separation of variables method to deal with nonlinear partial differential equations" at One week STTP on "Recent trends of Mathematics in Science & Technology (RTMST-2017)" during 2 – 6 January 2017 organized by Department of Applied Science and Humanities , GIDC Engineering college, Navsari.
9. Delivered an expert talk on "**Some Interesting Mathematical Expressions involving Pi(π) with Mathematica**" at two days National Seminar on "World of Mathematical Modeling" held on 12th and 13th April, 2017 at Institute of Advanced Research, Gandhinagar.
10. Delivered an expert talk on "**Solution of nonlinear partial differential equations using Generalized Separable Method**" at two days National Conference on MATHEMATICAL MODELING-MODERN APPROACHES (MMMA-2017) held during 13 – 14 October, 2017 at DIT University, Dehradun.
11. Delivered an expert talk on "**An extension of Method of Separation of variables to deal with nonlinear partial differential equations**" at "54th Annual Conference of Gujarat Ganit Mandal" during 1 – 3 November 2017 organized by Mahila Science College, Dhrol.

12. Delivered an expert talk on "**Lin Bairstow's Method**" at National Seminar on "Advanced Numerical Methods (NSANM-2017)" during 1-3 December 2017 organized by Department of Mathematics, School of Technology, PDPU, Gandhinagar.
13. Delivered an expert talk on "**Mathematical Modeling of Physical phenomena arising during Secondary oil recovery process**" at National Conference on "Applied Mathematical Sciences" (NCAMS-2018) during 14-15 April 2018 organized by Department of Mathematics, Gujarat University and Department of Applied Sciences & Humanities, Parul University, Vadodara.
14. Delivered a scholarly talk on "**Scope of research in the field of Fluid flow through porous media**" at Research Meet for Mathematics during 06-08 December 2018 organized by Department of Applied Sciences & Humanities, Parul Institute of Engineering & Technology, Parul University, Vadodara.
15. Delivered an expert talk on "**Generalized Separable Method for solving Nonlinear PDE and its application**" at International Conference on "New Frontiers in Mathematics" during 11-13 December 2018 sponsored by UGC, DST and NBHM, organized by School of Mathematics, Jiwaji University, Gwalior (M.P), India.
16. Delivered an expert lecture on "**Laplace Transform and its applications**" at FDP (WORKSHOP) on "Recent Trends In Engineering Mathematics With Special reference To AI, ANN AND FEM" during 12 – 13 October, 2019 organized by Sagar Institute Of Research, Technology & Science (SIRTS), Bhopal (M.P), India.
17. Delivered an expert talk on "**Mathematical Modelling and Analysis of Some Physical phenomena arising in fluid flow through porous media**" at 8th International conference and 24th Annual conference of Gwalior Academy of Mathematical Science (ICGAMS-2K19) organized by VIT, Bhopal during 13-15 December 2019.

18. Delivered an expert talk on "**Advanced Numerical Techniques**" at DST- GUJCOST sponsored one day seminar on "Role of Mathematics in Engineering & Technology" organized by SCET-Surat on 1st February 2020.
19. Delivered keynote address on **Importance of Research and Innovations in Science and Technology at** International Conference on Science, Engineering & Technological Innovation ICSETI - 2020 (E-Conference) during 24 & 25 October, 2020 organized by Research Culture Society & Kryvyi Rih National University, Ukraine.
20. Delivered keynote address at International Conference on Science, Engineering & Technological Innovation ICSETI - 2021 during 10 & 11 July, 2021 jointly organized by Kryvyi Rih National University and Technology Department, Research Culture Society sponsored by Scientific Research Association, Ukraine at Bangkok, Thailand (Online mode)
21. Delivered keynote address at Eurasian Conference on 'Science, Engineering & Technological Innovations' ECSETI - 2021 during 20 & 21 November, 2021 jointly organized by Kryvyi Rih National University and Technology Department, Research Culture Society sponsored by Scientific Research Association.
22. Delivered invited talk on **Mathematical Solution of fluid flow problems by using Extended Method of Separation of Variables** at 9th International Conference and 25th (Silver Jubilee) Annual Conference of Gwalior Academy of Mathematical Sciences [ICGAMS-2K22] during 29 September to 1st October 2022 organized by Department of Applied Sciences and Humanities, Pimpri Chinchwad college of Engineering, Nigdi, Pune.
23. Delivered Invited Lecture **Extension of Method of Separation of Variables for solving nonlinear PDE** at International Conference of The Mathematical Society-BHU on Recent Trends in Mathematical and Computational Sciences (RTMCS - 2023) during 3 – 5 February 2023 organized by CIMS, Institute of Science, Banaras Hindu University, Varanasi.

24. Delivered Invited Lecture **Mathematical Modeling of physical phenomena occurs during secondary oil recovery process** at one day workshop on “Real life applications of Mathematical and Physical Sciences (RLAMPS-2023) on 25th July 2023 organized by Department of Mathematics, SVNIT, Surat.
25. Delivered Invited Lecture **Mathematical Modeling of Fluid Dynamics in Porous Media** at International Conference on "Computational Modeling and Sustainable Energy", during 15-17 December, 2023 organized by PDEU, Gandhinagar.

BOOK PUBLICATION

1. **Title:** Introduction to Fluid flow through Porous media
   ISBN: 978-3-659-61845-1
   Publisher: Lambert Academic Publishing.
   Released on: 21st October 2014
2. **Title:** Solution of one dimensional Groundwater flow problems
   ISBN: 978-3-659-68619-1
   Publisher: Lambert Academic Publishing.
   Released on: 3rd February 2015

BOOK CHAPTER

1. Title of Book Chapter: “Differential Quadrature Method for Solving Linear and Non-Linear Differential Equation”

Edited Book titled: “RECENT ADVANCEMENT OF MATHEMATICS IN SCIENCE AND TECHNOLOGY”.

Authors: Jishan Shaikh, Amit K Parikh

ISBN NO: 978 – 81-950475-0-5

Publication date: MARCH, 2021

Publisher: JPS Scientific Publications, Tamil Nadu, India.

2.Title of Book Chapter: “A series solution of Nonlinear equation for Imbibition phenomenon in one dimensional Heterogeneous porous medium”

Edited Book titled: “Advances in Intelligent Systems and Computing”.

Authors: Dhara T Patel, Amit K Parikh

Proceedings of the First International Conference, MMCITRE 2020, AISC vol 1287, pp 29 - 40, Springer, Singapore. (https://doi.org/10.1007/978-981-15-9953-8_4), Feb 2021.

Print ISBN978-981-15-9952-1 Online ISBN978-981-15-9953-8

3.Title of Book Chapter: “*Numerical Solution of Counter-Current* Imbibition *Phenomenon In Homogeneous Porous Media Using Polynomial Base Differential Quadrature Method With Chebyshev-Gauss-Lobatto Grid Points*”

Edited Book titled: “Advances in Intelligent Systems and Computing”.

Authors: Amit K Parikh, Shaikh Jishan

Proceedings of the First International Conference, MMCITRE 2020, AISC vol 1287, pp 195 - 206, Springer, Singapore. https://doi.org/10.1007/978-981-15-9953-8_17), Feb 2021.

4.Title of Book Chapter: “Analysis of One-Dimensional Groundwater Recharge by Spreading using Hybrid Differential Transformation and Finite Difference Method”

Edited Book titled: “Computational and Analytic Methods in Biological Sciences: Bioinformatics with Machine Learning and Mathematical Modelling”.

Authors: Aruna Sharma, Amit K Parikh

Electronic ISBN NO: 9788770226943, Print ISBN: 9788770226950

Publication date: MARCH, 2023, PP. 247 - 259

Publisher: River Publication, IEEE Xplore

5.Title of Book Chapter: “Semi-analytic Numerical Methods to Solve One-Dimensional Dispersion Phenomenon of Miscible Fluid Flow Through Porous Media”

Edited Book titled: “Mathematical Modeling, Computational Intelligence Techniques and Renewable Energy” Proceedings of the Third International Conference, MMCITRE 2022

Book Series: Advances in Intelligent Systems and Computing, Volume - 1440

Authors: Aruna Sharma, Amit K Parikh

ISSN 2194-5357 ISSN 2194-5365 (electronic) ISBN 978-981-19-9905-5 ISBN 978-981-19-9906-2 (eBook) Publication date: MAY, 2023 PP. 401 - 414

https://doi.org/10.1007/978-981-19-9906-2

Publisher: Springer

STTP/SEMINAR/CONFERENCE/WORKSHOP ORGANIZED

1. One day seminar on 'Paper Presentation in Mathematics' organized on 28 August 2010 at Institute of Technology, Nirma University, Ahmedabad.
2. Two days national seminar on 'Mathematical Modelling and Simulation' organized during 3-4, May 2013 at Institute of Technology, Nirma University, Ahmedabad.
3. ISTE approved One week STTP on "Recent Trends in Application of Mathematics in Engineering and Science" (RTAMES-2013) organized during 1-5, July 2013 at Institute of Technology, Nirma University.
4. One day national conference on "Recent trends in Mathematics and its Applications" (RTMA – 2014) on 19th Dec 2014 at C.U.Shah University, Wadhwan city.
5. International Conference on Emerging Trends in Scientific Research (ICETSR-2015) during 17-18 December 2015 at C.U.Shah College of Engineering and Technology, Wadhwan city.
6. One day seminar on Recent Trends in Biological Sciences (RTBS-2016) on 11 March 2016 at Mehsana Urban Institute of Sciences, Ganpat University, Kherva.
7. CARS (Centre for Advanced Research Studies), Ganpat University sponsored A Short term Hands-on training program on "**Innovative approach in Mathematics Education"** during 08-12, Dec 2016 at Mehsana Urban Institute of Sciences, Ganpat University, Kherva.
8. "**53rd Annual conference of Gujarat Ganit Mandal**" during 23 – 25 December 2016 at Mehsana Urban Institute of Sciences, Ganpat University, Kherva.

9. **“3rd International Young Scientist Congress (IYSC-2017)”** and Workshop on **“Scientific Writing”** during 8th - 9th May 2017 at Mehsana Urban Institute of Sciences, Ganpat University, Mehsana, Gujarat, India in collaboration with International Science Community Association (ISCA).
10. 22nd National Cum 6th International Conference of the Gwalior Academy of Mathematical Sciences on **“Advances in Pure and Applied Mathematics”** and Symposium on **“Mathematical Methods and Models in Bio Sciences”** during 22-24 December 2017 at Mehsana Urban Institute of Sciences, Ganpat University, Mehsana, Gujarat, India.
11. National level online workshop on “Scientific Writing’ during 14-15 August 2021.
12. Organized a one day National Symposium on “ Recent Trends in Mathematical Sciences “ (RTMS 2022) on 30th April 2022 at Department of Mathematics, Ganpat University in association with **Fater Academy of India (FAI) & Gwalior Academy of Mathematical Sciences (GAMS).**
13. One week National Level online FDP on “ Recent Trends in Mathematics for Engineers & Scientists” during 25 -30 July 2022.
14. Organized **“A One-Day Workshop on Mathematical Models: Innovative Approach in Mathematics Education”** to honor and acknowledge the contributions of a renowned Mathematician Late Prof. P. C. Vaidya in association with Vigyan Gurjari on Saturday, 22nd July 2023
15. Organized अंतरिक्ष 2023: A one day hands on workshop on "Arduino-Based Circuits" as a part of celebration of the Birth Anniversary of Dr. Vikram A. Sarabhai, Father of Indian Space Program on 11 th August and 12 th August, 2023.
16. Organized an International conference on **Advances in Pure and Applied Mathematics (ICAPAM - 2023)** during 27 – 28 October 2023 at Department of Mathematics, Ganpat University.

STTP/SEMINAR/CONFERENCE/WORKSHOP ATTENDED

1. Attended Orientation Program from 14/05/2001 to 06/06/2001(including Sunday & Holidays) organized by Academic Staff Collage, Gujarat University at VallabhVidyanagar under UGC scheme.(4 weeks)

2. Participated in AICTE-ISTE sponsored STTP on “Applications of Integral Transforms Techniques in Science and Engineering” from 17/11/2003 to 28/11/2003 organized by Dr.Babasaheb Ambedkar Technological University, Lonere.(2 weeks)
3. Attended UNESCO & NBHM sponsored National Instructional Workshop on “Industrial Mathematics” from 01/12/2003 to 07/12/2003 organized by M.S.University of Baroda & IIT Bombay at Vadodara.(1 week)
4. Participated in AICTE sponsored Staff Development Program on “Enhancing Skills of Engineering Faculty” from 28/06/2004 to 10/06/2004 organized by Nirma University of Science & Technology, Ahmedabad. (2 weeks)
5. Attended Indo-Uk Study Meet on ‘Industrial Problems’ from 20/03/2006 to 24/03/2006 organized by Department of Applied Mathematics, M.S University, Vadodara.(1 week)
6. Attended QIP-STTP on ‘Harmonic Analysis and Partial Differential Equations’ from 01/01/2007 to 12/01/2007 organized by IIT Madras, Chennai. (2 weeks)
7. Participated in an International Symposium on ‘Harmonic Analysis, Wavelets and Partial Differential Equations’ on 8th January, 2007,organized by IIT Madras, Chennai. (1 Day)
8. Attended a Seminar on ‘Mathematics Meet 2008’ during 28-29 March 2008 sponsored by UGC & GUJCOST organized by Department of Mathematics, Gujarat University, Ahmedabad.(2 days)
9. Attended a QIP-STTP on ‘Role of Soft Skills in Effective Class room Teaching’ at IIT Roorkee from 23-06-08 to 04-07-08.(2 weeks)
10. Attended an STTP on “Mathematical Models for Engineers” from 25-05-09 To 29-05-09 organized by Maths & Humanities Dept., Institute of Technology, Nirma University, Ahmedabad. (1 week).
11. Attended 55th Congress of ISTAM (An International meet) during 18-21,Dec 2010 organized by NIT Hamirpur.

12. Attended 76th Annual Conference of The Indian Mathematical Society during 27-30,Dec 2010 organized by SVNIT Surat.
13. Attended 16thAnnual cum 2ndInternational Conference of Gwalior Academy of Mathematical Sciences during 22-25, Sept 2011organized by S.S.Dempo College of Commerce and Economics, Panaji, Goa.
14. Attended 56th congress of Indian Society of Theoretical and Applied Mechanics(ISTAM), (An International meet) during 19-21, Dec 2011organized by SVNIT Surat.
15. Attended 'Science Academies' lecture Workshop on Partial Differential Equation and its Applications during 1-4, March 2012 organized by SVNIT Surat.
16. Attended one week STTP on 'Mathematical Methods for Physical Sciences and Engineering' during 6-10, May 2013 organized by SVNIT Surat.(1 week)
17. Attended one week TEQIP-II sponsored STTP on 'Advanced Analytical & Numerical Techniques for Engineers and Scientists' during 3-7, March 2014 organized by SVNIT Surat. (1 week)
18. Attended one day FDP on 'Soft Skills Development' on 18th July 2015 organized by C.U. Shah College of Engineering & Technology, Wadhwancity.
19. Successfully completed online course on "Technical Communication for Scientist and Engineers" as a part of blended STTP from the project of NMEICT (MHRD), offered by IIT Bombay during Nov-Dec 2015.
20. Attended one day workshop on "Revised Accreditation Framework of NAAC" on 13th February, 2019 organized by Knowledge Consortium of Gujarat & NAAC, Bangalore.
21. Attended 2nd International conference on Mathematical Modelling and Simulation in Physical Sciences (MMSPS - 2022) organized by SVNIT Surat during 5 -6 February, 2022 (online mode).

AWARDS / RECOGNITIONS ACHIEVEMENTS

- Achieved 2nd rank in Srinivasa Ramanujan Mathematics Award Examination – 2012 at State level conducted by ISTE and got the cash prize of Rs. 2000/-
- Consolation prize at national level in Srinivasa Ramanujan Mathematics Competition for engineering teachers – 2012 conducted by ISTE and got the cash prize of Rs. 5000/-
- Achieved 2nd rank in Srinivasa Ramanujan Mathematics Award Examination – 2013 at State level conducted by ISTE and got the cash prize of Rs. 2000/-
- Working as a reviewer in World Journal of Social Science, ISSN 2329-9347(Print), ISSN 2329-9355 (online), Sciedu Press, USA.
- PhD RDC/DPC expert for GTU, Rai University, Carolx Teacher's University, C.U.Shah University
- Received certificate of appreciation from "International Forum for Excellence in Higher Education" for successfully conducting "National Creativity Aptitude Test 2015" at C.U.Shah College of Engineering & Technology, Wadhwancity.
- Received certificate of appreciation from "International Forum for Excellence in Higher Education" for successfully conducting "National Creativity Aptitude Test 2016" at Mehsana Urban Institute of Sciences, Ganpat University
- Obtained Certificates of Honour for publishing and presenting research papers in the year 2010-11, 2011-12 and 2012-13 on occasion of Foundation day celebration of Institute of Technology, Nirma University, Ahmedabad.
- Worked as an editor for Souvenir of 53rd Gujarat Ganit Mandal Adhivation **"Smarnika"** organised by Mehsana Urban Institute of Sciences during 23-25 December 2016.
- Received national level award for 'Excellence in Education' from Confederation of Education Excellence (CEE), New Delhi on 22nd April 2017.

- Received **“Education Leadership Award”** from **Dewang Mehta national education awards (Regional round) & 25th Business School Affairs** as a celebration of 25th Silver Jubilee year in October 2017.
- The Executive council of **Gwalior Academy of Mathematical Sciences** nominated me as Vice President of GAMS for three years from 1st April 2018.
- Elected as an **Associate Fellow** by “Gujarat Science Academy (GSA)” *w.e.f* January 2018.
- Worked as a Managing editor for Special Issue of **Mathematics Today, Volume: 34A, April 2018, [**ISSN: 0976-3228, e-ISSN: 2455 – 9601]
- Received **Best Paper Award** for research paper titled “*Mathematical modeling of physical phenomena arising during oil recovery process”*, at **68th International Conference on New Trends in Engineering, Science and Management** during 20-21 July 2018, organized by IOSRD at Visakhapatnam.
- Received “**Life Time Achievement Award**” in International Conference “IOSRD Research Awards” during 20-21 July 2018 at Visakhapatnam organised by **International Organisation of Scientific Research and Development (IOSRD).**
- Invited as Judge in “Science Excellence 2018” to evaluate Poster & Oral presentations for the mathematics subject on 20th September 2018 organised by Gujarat University.
- Invited as Judge in “Mathematics Excellence 2019 (MathXL-2019)” on 5th January 2019 organised by Department of Mathematics, Gujarat University.
- Received **I2OR National Eminent Researcher Award 2020** from International Institute of Organized Research in November 2020.
- Received Ganpat University President Award for Manager of the Year for the year 2019 with cash prize of INR 2,00,000/-. This award was presented on Vidyaya Samajotkarsh Divas 12th January 2021 to recognize the extraordinary performance in the area of Institutional Capacity building, excellency in teaching & learning, creating enabling culture,

excellent admission & placement, sustainable growth and innovative initiatives which does place the institute in premium category of Education.

- Received **Outstanding Scientist Award** at International Scientist Awards on Engineering, Science and Medicine during 06-07 March 2021, Goa, India organised by VDGOOD Professional Association.
- Received **Research Excellence Award 2021** from InSc (Institute of Scholars) for the publication *Mathematical Solution of Fingering Phenomenon In Vertical Downward Direction Through Heterogeneous Porous Medium,* Advances in Mathematics: Scientific Journal 10 (2021), no.1, 483–496 ISSN: 1857-8365 (printed); 1857-8438 (electronic), Spec. Issue on ICIRPS-2020 [SCOPUS Indexed] in April 2021.
- Received **Excellence Research Award** INSO – 2023 at **International Conference for Award Winners in Engineering, Science and Medicine** held on **15-Apr-2023** in **Pondicherry**, India.
- Received **Best PG Science Teacher Award 2022** in Mathematical Sciences (for age category < 55 years) from **Gujarat Science Academy** (GSA)

Area of Interest:

- Fluid flow through porous media
- Numerical Analysis
- Differential Equations
- Discrete Mathematics
- Mathematical Modelling
- Linear Algebra
- Integral Transform
- Complex Analysis

RESEARCH STATEMENT

The topic of my M.Phil Dissertation was **'Study of Continuous Functions'** wherein the notions of continuity and related topics had been studied. In this research work, continuity and its equivalent formulations had been derived in real line, metric space and topological spaces. I have studied some important properties and theorems on continuity along with the notions of compactness, connectedness, mean value theorems, Riemann integration and graph. The classical as well as the generalized version proofs of one of the most interesting result of analysis namely 'Ascoli's Theorem' had been given in the dissertation.

My Ph.d thesis titled **'Analytical Solution of Nonlinear Partial Differential Equations of some phenomena arising in fluid flow through porous media'** is the description of my research work carried out on analytical solution of the problems of fluid flow through porous media. In this research work, different phenomena arising during secondary oil recovery process have been studied. Mathematical formulation of the problems leading to nonlinear partial differential equations and the solutions have been obtained by using different methods such as perturbation method, similarity transformation and generalized separable method.
This research work carried out is very useful to oil industry.

In my Ph.d thesis, different physical phenomena arising during secondary oil recovery process have been discussed in homogeneous porous medium in horizontal direction as well as in vertical downward direction.

According to my future research plan, this study can be extended for two dimensions and three dimensions. For multiphase flow problems in displacement process, only two immiscible phases have been considered but in future it may be studied for more than two phases where the simultaneous flow of oil, water and gas may be considered. In different physical phenomena, the external effects such as gravitational effect and magnetic field effect have been considered but other external effects such as temperature, tsunami and earthquake which affect the underground flow system may be considered in future work. In double phase flow problems, two immiscible fluids oil and water have been considered but in future two miscible fluids oil and solvent (methane/propane/carbon dioxide) in oil recovery problems may be considered. In this research work, it is assumed that the porous medium is homogeneous in which rock properties of porosity and permeability are considered as constants but in reality the medium is heterogeneous so all the problems can be developed in heterogeneous porous media where rock properties of porosity and permeability are functions of variables.

For the solution of governing non linear partial differential equations, the advanced numerical mathematical methods such as finite element method, finite volume method, differential quadrature method, homotopy perturbation method, variational iteration method may be applied.

A wide range of applications can be studied as future projection in the area of convective heat transfer in fluid-saturated porous media such as thermal insulation engineering, water movements in geothermal reservoirs, heat pipes, underground spreading of chemical waste, nuclear waste repository, geothermal engineering, grain storage and enhanced recovery of petroleum reservoirs. Radiative heat transfer and multiphase transport processes in porous media, both with and without phase change, have gained extensive attention in recent years. This is due to the wide range of applicability of these research areas in contemporary technology. These applications include, but are not restricted to, areas such as geothermal engineering, building thermal insulation, chemical catalytic reactors, packed cryogenic microsphere insulation, petroleum reservoirs, direct contact heat exchangers, coal combustors, nuclear waste repositories, and heat pipe technology.

Several applications related to porous media require a detailed analysis of convective heat transfer in different geometrical shapes, orientations and configurations. Based on the specific applications, the flow in the porous medium may be internal or external. Most of the studies in porous media carried out until the past two decades are based on the Darcy flow model, which in turn is based on the assumption of creeping flow through an infinitely extended uniform medium. However, non-Darcian effects are quite important for certain applications. Different models can be introduced for studying and accounting for non-Darcian effects such as the inertial, boundary, and variable porosity effects.

Many industrial operations in the areas of chemical and metallurgical engineering involve the passage of a fluid stream through a packed bed of particulate solids to obtain extended solid fluid interfacial areas or good fluid mixing. Typical examples of applications involving such systems include catalytic and chromatographic reactions, packed absorption and distillation towers, ion exchange columns, packed filters, pebble-type heat exchanger, petroleum reservoirs, geothermal operations and many others. The design of these systems is decided by mechanisms of pressure drop, fluid flow and heat and mass transfer governing the process in the packed bed arrangement. Considerable attention has been paid to the aforementioned aspects because of their direct influence on the optimization and stability of the design of these systems.

Developments in modeling transport phenomena in porous media have advanced several pertinent areas, such as bio transport in porous media, turbulent modeling in porous media, etc.

My areas of interest for the research are Fluid flow through porous media, Mathematical modeling, Numerical Analysis, and Differential equations.

List of References

1. Prof. Nita H.Shah
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